Training-Size Conformal Sweep for Zoo Uncertainty Quantification

Quick Technical Writeup

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Abstract

This document reports a training-size sweep for uncertainty quantification in NFL rushing-yard prediction. Three probabilistic models (Ridge, gradient-boosted trees, and a distributional CNN) are evaluated under a fixed game-level partition, and their predictive distributions are converted to conformal intervals via locally weighted calibration. The analysis quantifies how interval sharpness, empirical coverage, and CRPS change as the number of training games increases.

1 Objective

The inferential target is a calibrated prediction interval for an ordinal rushing-yard class, with nominal coverage $1 - \alpha = 0.90$ ($\alpha = 0.10$). The study evaluates whether additional training data improve:

- **Distribution quality**, measured by CRPS.
- **Interval sharpness**, measured by mean interval width.
- **Calibration**, measured by empirical coverage on a fixed test set.

2 Methodology

2.1 Data Partition and Sweep Design

All samples are assigned by game to one of three disjoint partitions: training pool (60%), calibration (20%), and test (20%). The calibration and test partitions are fixed once and held constant throughout the sweep.

Within the training pool, games are ordered by a fixed random permutation. Sweep conditions are nested prefixes of this ordering in 20-game increments (20, 40, ..., 360 games in the present run). Consequently, each condition differs only in training-set size; all other components of evaluation are invariant.

2.2 Model-Fitting Protocol

Three models are trained at each sweep size:

- Multinomial Ridge classifier.
- Multiclass gradient-boosted tree classifier.
- Distributional CNN (Zoo), optimized with CRPS-oriented training and early stopping.

For the tabular baselines, hyperparameters are tuned once on a dedicated tuning holdout (40 games drawn from the training pool) and then fixed for all sweep points. This prevents confounding between training-size effects and repeated retuning.

2.3 Conformal Interval Construction

Each model outputs a predictive mass function $\hat{p}_x(c)$ over 80 yardage classes, with cumulative distribution

$$\hat{F}_x(c) = \sum_{j \le c} \hat{p}_x(j).$$

A central model interval is first formed via predictive quantiles:

$$\ell(x) = \min\{c : \hat{F}_x(c) \ge \alpha/2\}, \qquad u(x) = \min\{c : \hat{F}_x(c) \ge 1 - \alpha/2\}.$$

On calibration samples (x_i, y_i), nonconformity is the minimum symmetric padding required to include the observed class:

$$s_i = \max\{\ell(x_i) - y_i,\ y_i - u(x_i),\ 0\}.$$

To obtain locally adaptive conformalization, each sample is represented by distributional locality features

$$\phi(x) = (\mu(x), \sigma(x), H(x)),$$

where μ, σ, and H are predictive mean, predictive standard deviation, and predictive entropy. For a test point x, calibration scores are weighted by a Gaussian kernel in standardized feature space:

$$w_i(x) = \exp\left(-\frac{\|\tilde{\phi}(x_i) - \tilde{\phi}(x)\|_2^2}{2h(x)^2}\right),$$

with adaptive bandwidth $h(x)$ equal to the distance to the k-th nearest calibration point ($k = 200$). The local padding is the finite-sample corrected weighted quantile:

$$q(x) = \left\lceil Q^{w(x)}_{(1-\alpha)(n+1)/n}\big(\{s_i\}_{i=1}^n\big) \right\rceil.$$

The final conformal interval is

$$L(x) = \max\{0, \ell(x) - q(x)\}, \qquad U(x) = \min\{79, u(x) + q(x)\}.$$

2.4 Reported Metrics

For each model and sweep size, we report:

- Test CRPS.
- Empirical coverage $\mathbb{P}(y \in [L(x), U(x)])$.
- Mean index-gap width $U(x) - L(x)$ and inclusive-bin width $U(x) - L(x) + 1$.

3 Results

Table 1 reports performance at the first and final sweep points (20 and 360 training games).

Table 1: Training-size sweep summary. Width denotes index-gap width ($U - L$).

Model	CRPS@20	CRPS@360	Width@20	Width@360	Mean Coverage
Ridge	0.01458	0.01449	21.40	21.60	0.9362
Tree	0.01444	0.01377	15.96	13.82	0.9143
Zoo CNN	0.01600	0.01327	20.29	13.58	0.9163

Key findings:

- The distributional CNN exhibits the largest CRPS reduction (0.01600 $\rightarrow$ 0.01327) and substantial interval narrowing (20.29 $\rightarrow$ 13.58).
- The tree baseline also improves in both CRPS (0.01444 $\rightarrow$ 0.01377) and width (15.96 $\rightarrow$ 13.82), though less than the CNN at larger sample sizes.
- The Ridge baseline remains comparatively flat in CRPS and maintains wider intervals.
- All methods remain conservative with respect to nominal 0.90 coverage (approximately 0.91–0.94).

4 Interpretation

The sweep indicates that increasing sample size primarily benefits non-linear distributional models in this problem. The locally weighted conformal step preserves valid, conservative coverage while allowing interval width to contract in regions where the predictive distribution is more concentrated.

5 Conclusion

In this experiment, larger training sets systematically improve predictive quality and interval sharpness for the non-linear models, most notably the distributional CNN. At the same time, conformal calibration maintains empirical coverage above the nominal 90% target across all sweep points.

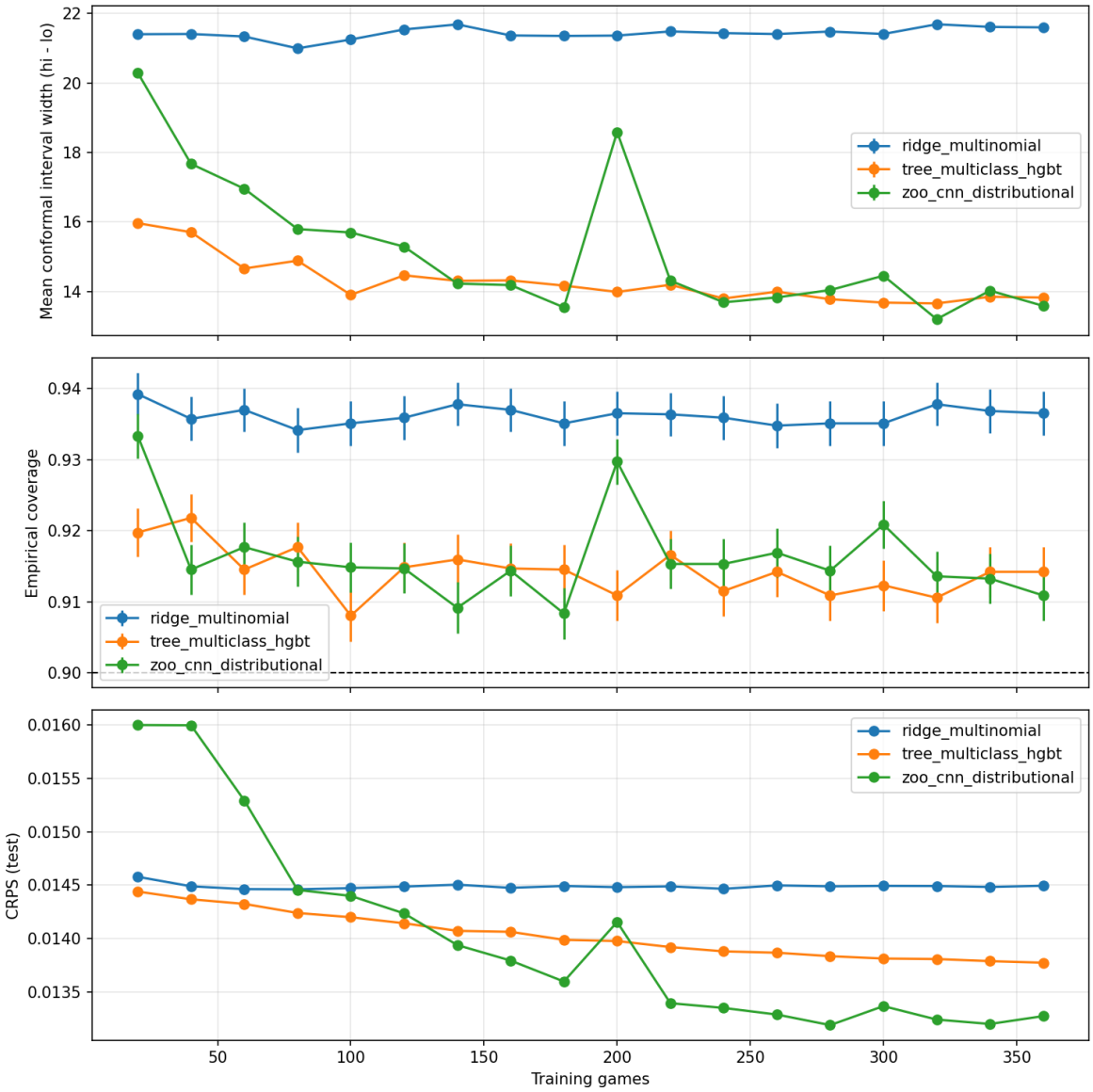

Figure 1: Interval width, empirical coverage (dashed line at 0.90), and CRPS as functions of training-set size.